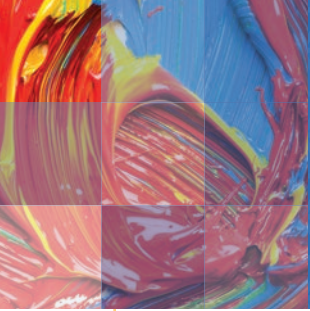




Inequalities in Two Variables

When the equals sign in a linear equation is replaced by an inequality symbol, the statement becomes a linear inequality.

Definition

A **linear inequality** uses terms with degree zero or one and an inequality symbol to relate two variables.

The graph of a linear inequality is a region of the coordinate plane. The region is determined by a line, called the boundary line. The boundary line is the graph of the related equation, which uses an equals sign.

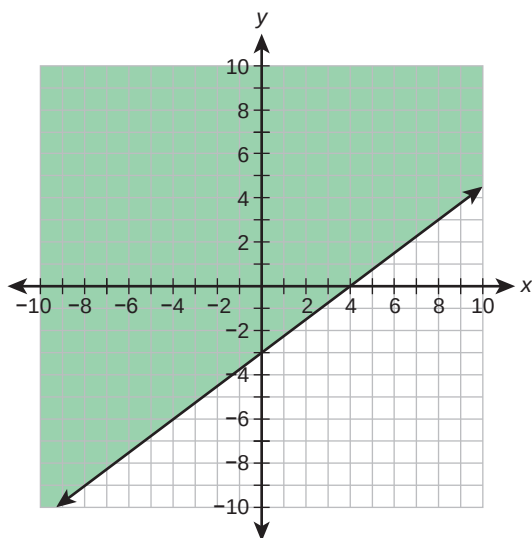
$$\text{linear inequality: } y \geq \frac{3}{4}x - 3$$

$$\text{related linear equation: } y = \frac{3}{4}x - 3$$

The boundary line divides the plane into two **half-planes**. If the inequality is a nonstrict inequality (that is, it uses $\leq$ or $\geq$), then the boundary line is a solid line and the half-plane is a **closed half-plane**. If the inequality is a strict inequality (it uses $<$ or $>$), then the boundary line is a dashed line and the half-plane is an **open half-plane**.

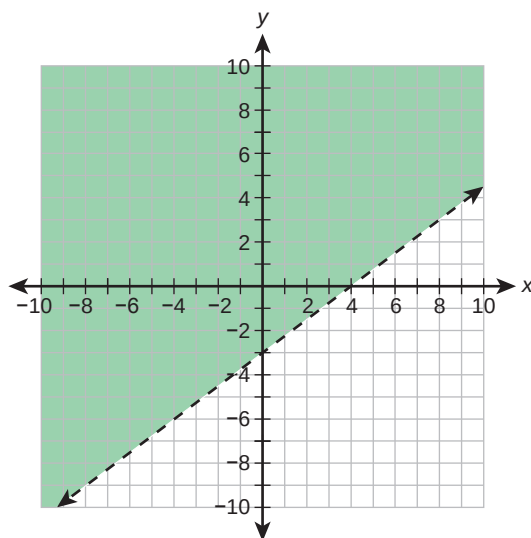
Closed Half-Plane

$$y \geq \frac{3}{4}x - 3$$



Open Half-Plane

$$y > \frac{3}{4}x - 3$$



To determine which half-plane to shade, use a test point. You can choose any point that is not on the boundary line.

When the inequality is written in the form such that the related equation is in slope-intercept form, then the region to be shaded may be determined without using a test point. For $>$ or $\geq$, shade above the boundary line (where the y -values on the y -axis are greater than the y -intercept). For $<$ or $\leq$, shade below the boundary line (where the y -values on the y -axis are less than the y -intercept).

Any point in the shaded region is a solution of the inequality. A point on a solid boundary line is also a solution, but a point on a dashed boundary line is not a solution.

► **Think About It** The point $(0, 0)$ is usually a good test point because it makes calculations simple.

Using the y-Intercept to Graph an Inequality in Two Variables

EXAMPLE 1

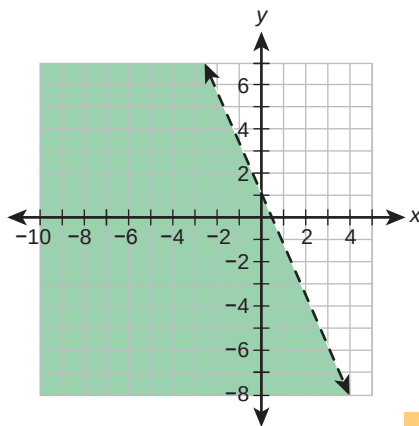
Graph $y < -2x + 1$.

SOLUTION

Step 1 Graph the related equation, $y = -2x + 1$. The y -intercept is 1, so first plot $(0, 1)$. Use the slope of -2 to find another point on the line by moving down 2 units and right 1 unit from that point.

Step 2 Draw a dashed line through the points because $<$ indicates that the boundary line does not contain solutions.

Step 3 Since the related linear equation is written in slope-intercept form and the inequality symbol is $<$, shade below the boundary line.



Graphing a Linear Inequality with One Variable

EXAMPLE 2

Graph $x \geq 1.5$.

SOLUTION

Step 1 Graph the related equation, $x = 1.5$. The line is a vertical line passing through $(1.5, 0)$.

Step 2 Use a solid line because $\geq$ indicates that the boundary line contains solutions.

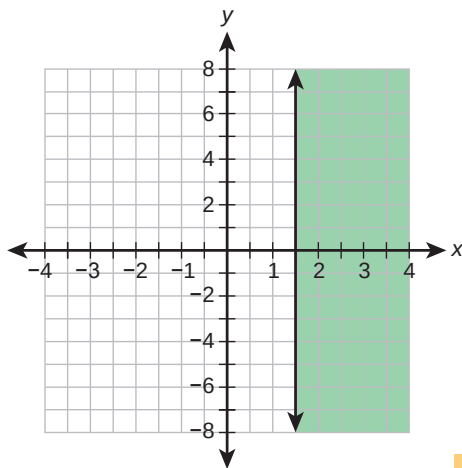
Step 3 Choose a test point to determine which half-plane to shade.

test-point: $(0, 0)$

$$x \geq 1.5$$

$$0 \not\geq 1.5$$

The point $(0, 0)$ is not in the solution set; the half-plane that does not contain the origin is shaded. Shade to the right of the line.



Using Intercepts to Graph an Inequality in Two Variables

EXAMPLE 3

Graph $2x - 5y > 10$.

SOLUTION

Step 1 Use intercepts to find two points on the graph of the related equation.

$$\begin{aligned}2x - 5y &= 10 \\2 \bullet \textcolor{red}{0} - 5y &= 10 \\-5y &= 10 \\y &= -2\end{aligned}$$

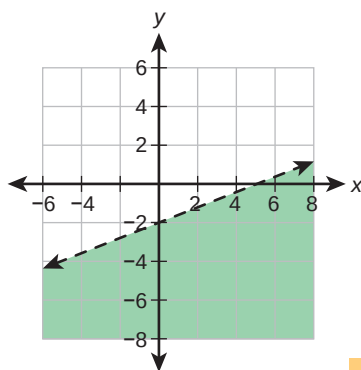
$$\begin{aligned}2x - 5y &= 10 \\2x - 5 \bullet \textcolor{blue}{0} &= 10 \\2x &= 10 \\x &= 5\end{aligned}$$

Step 2 Draw a dashed boundary line through $(0, -2)$ and $(5, 0)$.

Step 3 Choose a test point to determine which half-plane to shade.

$$\begin{aligned}2x - 5y &\stackrel{?}{>} 10 \\2 \bullet \textcolor{red}{0} - 5 \bullet \textcolor{blue}{0} &\stackrel{?}{>} 10 \\0 &\nlessgtr 10\end{aligned}$$

The point $(0, 0)$ is not in the solution set, so the half-plane that does not contain the origin is shaded.

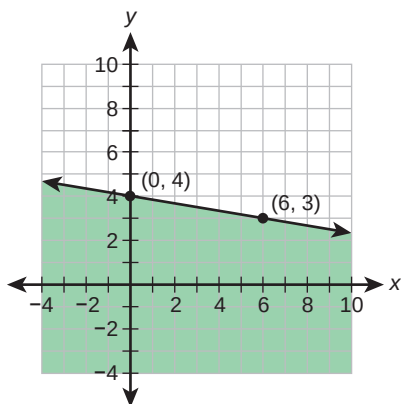


Writing Linear Inequalities from Graphs

EXAMPLE 4

Write an inequality for the graph.

A



SOLUTION

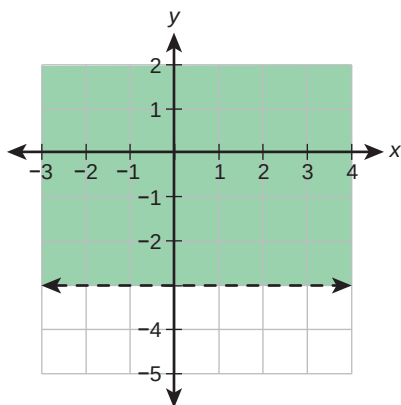
Step 1 Determine the equation of the boundary line. The y -intercept is 4 and the slope is $-\frac{1}{6}$. The equation of the boundary line is $y = -\frac{1}{6}x + 4$.

Step 2 Determine the inequality sign. Because the boundary line is solid, the inequality will be a nonstrict inequality, using either $\leq$ or $\geq$. Because the shading is below the line, use $\leq$.

Step 3 Write the inequality.

$$y \leq -\frac{1}{6}x + 4$$

► **Think About It** The point $(0, 0)$ satisfies $y \leq -\frac{1}{6}x + 4$ because $0 \leq 4$.

B**SOLUTION**

Step 1 Determine the equation of the boundary line. The y -value of every ordered pair is -3 , so the equation is $y = -3$.

Step 2 Determine the inequality sign. Because the boundary line is dashed, the inequality will be a strict inequality, using either $<$ or $>$. Because the shading is above the line, use $>$.

Step 3 Write the inequality.

$$y > -3 \quad \blacksquare$$

► **Remember** The slope of a horizontal line is 0. The equation of the boundary line can be written as $y = 0x - 3$.